360° Street View Integration

GitHub Copilot implementation prompt for an existing Angular 21 + ArcGIS Maps SDK application

Project context

I have an existing Angular 21 Smart Banking AI Information System. My application already uses Angular 21, ArcGIS Maps SDK for JavaScript, ArcGIS MapView, existing GIS search and zoom functionality, a vertical left-side toolbar, and a green corporate theme. I want to integrate Google Street View 360° without replacing or redesigning my existing GIS application.

1. Required workflow

1. The user searches for any location using the existing search functionality (for example Riyadh, Al Malaz, Olaya, King Fahd Road, Diplomatic Quarter, an ATM location or a bank branch).

2. The existing ArcGIS functionality finds the selected location, extracts its latitude and longitude, zooms to it, and displays it on the map.

3. The user clicks a new dedicated **360° Street View** button in the existing left vertical GIS toolbar.

4. The application retrieves the currently selected location coordinates, calls Google StreetViewService, finds the nearest available panorama, and displays Google StreetViewPanorama in a large panel above the ArcGIS map.

5. The ArcGIS map remains available underneath. The user controls when Street View opens; it must not open automatically after a search.

Use the selected search result coordinates, never the user's current GPS location.

2. Viewer layout

Create a large expandable/collapsible 360° viewer. Suggested initial layout: viewer approximately 45% of available height and ArcGIS map approximately 55%.

Provide close, full-screen, zoom, 360° rotation, current panorama location, and imagery-date controls (when available).

When closed, return to the original full-size ArcGIS map and preserve its map extent, selected layers and graphics.

3. Google Street View integration

Use the official Google Maps JavaScript API with **google.maps.StreetViewService** and **google.maps.StreetViewPanorama**.

Search for nearby panoramas using the selected ArcGIS coordinates. Start with a 100-metre radius; if unavailable, retry at 250 metres and then 500 metres.

Display the actual distance between the selected location and the panorama. Do not present a distant panorama as if it were at the searched location.

If imagery is unavailable, display: **“Street View imagery is not available near this location.”**

4. Button behaviour

If no location has been selected, display: **“Please search and select a location first.”**

If a location is selected, open the Street View viewer and search for its nearest panorama.

When another location is selected, update the selected coordinates. Clicking 360° View again must show the panorama for the newly selected location.

If the viewer is already open, update the existing panorama instance instead of creating another one.

5. Angular architecture

Inspect the existing project before creating files. Create or reuse the following structure, adapting it to the current architecture:

```
src/app/services/google-maps-loader.service.ts
src/app/services/google-street-view.service.ts
src/app/components/street-view/street-view.component.ts
src/app/components/street-view/street-view.component.html
src/app/components/street-view/street-view.component.scss
```

6. Integration with existing ArcGIS search

Identify the existing search component and ArcGIS MapView component. Reuse the selected search result's latitude and longitude.

Maintain a shared selected-location state using the existing state-management approach. If none exists, use an Angular service or signals.

Example data structure (illustrative only; use actual coordinates from the selected result):

```
{
  name: 'Al Malaz',
  latitude: 24.65,
  longitude: 46.72
}
```

The example coordinates above are placeholders, not production coordinates.

7. Visual design

Follow the existing Smart Banking GIS interface: corporate green, light-green gradients, beige backgrounds, rounded toolbar buttons and a professional enterprise GIS appearance.

Place the 360° button in the existing left vertical toolbar without removing or changing existing buttons. Add tooltip: **“Open 360° Street View”**.

8. Technical requirements

Load Google Maps API only once; reuse the StreetViewPanorama instance; handle API loading errors and missing imagery; clean up event listeners; avoid unnecessary API calls; preserve the existing ArcGIS MapView instance; do not interfere with search, layers, popups or dashboards.

Keep the Google API key in environment configuration. Apply Google Maps API key restrictions and billing configuration.

9. Final testing

1. Open the existing GIS application.
2. Search for Al Malaz.
3. Confirm ArcGIS zooms to Al Malaz.
4. Click the new 360° Street View button.
5. Confirm the top panorama viewer opens.
6. Rotate the panorama.
7. Close the panorama.
8. Search for another location.
9. Click 360° Street View again.
10. Confirm the new panorama corresponds to the new search location.

10. Final expected result

Keep the existing GIS dashboard as it is. The only major UI addition should be a new 360° Street View button in the left vertical toolbar. The user controls when to open Street View.

Search → Zoom to location → Click 360° View → Display Google Street View panorama above ArcGIS map.

Implement this functionality without breaking existing Angular 21 or ArcGIS JS SDK features.

Implementation instruction for Copilot: Before modifying files, inspect the existing project structure, identify how the current search returns coordinates, and reuse existing services and interfaces wherever possible. Do not create duplicate map or search functionality. Implement incrementally and verify the existing dashboard still works.